

Hieu Nguyen

hieu7337@gmail.com | [Portfolio](#) | [LinkedIn](#) | [GitHub](#)

Summary

Engineer shipping and operating whole systems front to back, keeping them deliberately small. At Plenarius, a Go runtime serves 96 server-rendered pages and 205 public `/api/` paths from one binary. The architecture prioritises restraint: seven direct dependencies, no ORM, and React as non-hydrating islands rather than a single-page application. Prior production engineering at FPT Software and Viettel Post. Completing a Master of Artificial Intelligence at QUT, finishing November 2026.

Skills

- **Backend and frontend:** Go, Python, SQL, PostgreSQL, MongoDB, REST API design, server-rendered templates, job queues, background workers, FastAPI, React islands, esbuild, no-framework JavaScript.
- **Delivery:** Docker, CI/CD, generated configuration with reproducibility checks, migrations, deployment and operation of live services.
- **Cloud & infrastructure:** Azure (Functions, Cosmos DB), AWS (EC2, S3), Hetzner, Cloudflare (WAF, R2 object storage), Caddy, systemd.

Experience

Chief Technology Officer

Apr 2026 – Present

Plenarius (formerly Calimac)

- Consolidated production onto a single Go binary serving 96 server-rendered pages and 205 public API paths without a web framework, leaving zero PHP in the production request path.
- Delivered the interface as server-rendered pages with isolated client-side islands instead of a single-page application, supported by an asset build requiring no separate JavaScript toolchain.
- Operated the entire platform on a single commodity server for roughly US\$43 a month, maintaining reliability through a comprehensive automated test estate and strict continuous integration checks.

Research Assistant

Sep 2025 – Present

Queensland University of Technology

- Evaluated and engineered an agentic person re-identification system combining appearance, body composition, gait and physical attributes through confidence-weighted voting.
- Cleaned and prepared biomedical literature datasets, then fine-tuned transformer models for protein-protein interaction relation extraction.

AI Software Engineer

Apr 2023 – Nov 2024

Viettel Post

- Integrated Whisper fine-tuned on Vietnamese with BERT classifiers for offensive-language detection and mood prediction, achieving 90% accuracy in production.
- Trained an Extra Trees classifier for customer churn prediction, reaching 80% accuracy.

AI Software Engineer

Oct 2021 – Mar 2023

FPT Software

- Contributed to FPT's [US-patented](#) Sound AI acoustic anomaly-detection platform, deployed on a Samsung factory floor. Built its Python backend on Azure Functions, supporting model serving, CI/CD and production monitoring.

Research Intern, Computer Vision

Apr 2021 – Jul 2021

Asilla

- Fine-tuned and deployed real-time [human pose estimation](#) on Jetson Nano with TensorRT, using INT8 quantisation and structured pruning to improve efficiency by 20% under edge-compute con-

straints.

Projects & Awards

Local AI Workflow Platform

Presented | AgentCamp 2026, Brisbane Edition

Stack: Python, LangGraph, FastAPI, React Flow, Pydantic, YAML

Local-first platform for authoring, running and visualising AI workflows from declarative YAML. Each workflow is validated through a Pydantic GraphSpec contract before being compiled into runnable graphs, with approval interrupts, rollback restore and eval coverage.

Capture the Narrative

1st Place | UNSW National AI Competition

Stack: LLMs, multi-agent simulation, AWS EC2, S3

Built generative-AI agents inside a simulated election environment as an adversarial exercise in how synthetic content shifts sentiment.

SafeMind

Most Viable | TANDA GenAI Hackathon

Stack: Python, Flask, OpenAI API, MongoDB, python-docx

Voice-driven workplace incident capture that pre-fills WorkCover claim forms, compressing hours of paperwork for injured workers.

Education

Master of Artificial Intelligence

2025 – 2026

Queensland University of Technology | GPA 6.54/7.0

Bachelor of Information Technology

2018 – 2021

University of Science and Technology of Hanoi

Volunteer

Vice President

Oct 2025 – Present

QUT The Emerging Coders Hub (TECH)

Ran coding and IT workshops teaching beginners to build chatbots, personal websites and AI agents, and organised career events including mock interview sessions, progressing from General Executive (Jun–Oct 2025).

Project Manager

Sep 2025 – Present

QUT AI & ML Society

Led 30+ members building a predictive analytics dashboard on real insurance data for claim-delay prediction and anomaly detection, stepping up from Project Officer (Aug–Sep 2025).